

Case 2: Low-Dimensional Representation and Latent State Discovery in Wearable Biosignals

02582 Computational Data Analysis
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Abstract

This report investigates whether physiological features from the EmoPairCompete dataset reveal subject-invariant emotional or stress-related states. We use the provided heart rate, skin temperature, and electrodermal activity features and evaluate three preprocessing choices: global standardization, subject-wise normalization, and subtraction of the pre-puzzle resting baseline. PCA is used to derive low-dimensional representations, and KMeans clustering tests whether latent clusters align with the three experimental phases. The main finding is a negative but informative result: the raw representation is dominated by individual and cohort effects, while phase explains only a small part of the variance. Subject-wise normalization increases phase-related structure, but unsupervised clustering still does not reliably recover the experimental phases.

Research Questions

We ask whether the dominant directions of variation in the biosignal features are related to experimental phase or mainly driven by individual and cohort effects; whether subject-wise normalization or baseline subtraction improves the phase-related structure; and whether unsupervised clusters in a low-dimensional space correspond to pre-puzzle rest, puzzle task, and post-puzzle rest.

Data

We used the updated feature file HR_data_2.csv. It contains 312 observations, corresponding to 26 individuals, four rounds, and three phases per round. The input feature set contains 51 physiological features from HR, TEMP, and EDA. Metadata and questionnaire variables such as Phase, Individual, Cohort, Puzzler, and Frustrated were not used as model inputs; they were used only for interpretation and validation. The feature matrix contained only two missing values, which were median-imputed.

Method

Let X be the 312 by 51 physiological feature matrix. We compared three preprocessing pipelines. Raw scaled features use median imputation followed by global standardization. Subject-wise normalization centers and scales each feature within each participant, removing static physiological baselines. Phase1 baseline subtraction represents each observation as a deviation from the corresponding pre-puzzle resting observation within the same individual and round.

PCA was used to obtain a low-dimensional representation. The first five principal components were used for downstream evaluation. To diagnose what drives the representation, we computed a weighted R^2 between the PC1-PC5 scores and four metadata variables: phase, individual, cohort, and puzzler role. KMeans with $k=3$ was then applied to PC1-PC5, and cluster agreement with the known phases was measured using adjusted Rand index, normalized mutual information, silhouette score, and a cluster-by-phase heatmap.

As sanity checks, we also tested a one-class SVM trained only on phase1 resting data under leave-one-subject-out validation, and a supervised logistic regression classifier for phase2 versus phase1 under the same validation design. These checks were not the main unsupervised solution; they were used to assess whether any generalizable phase signal exists.

Results: PCA Structure

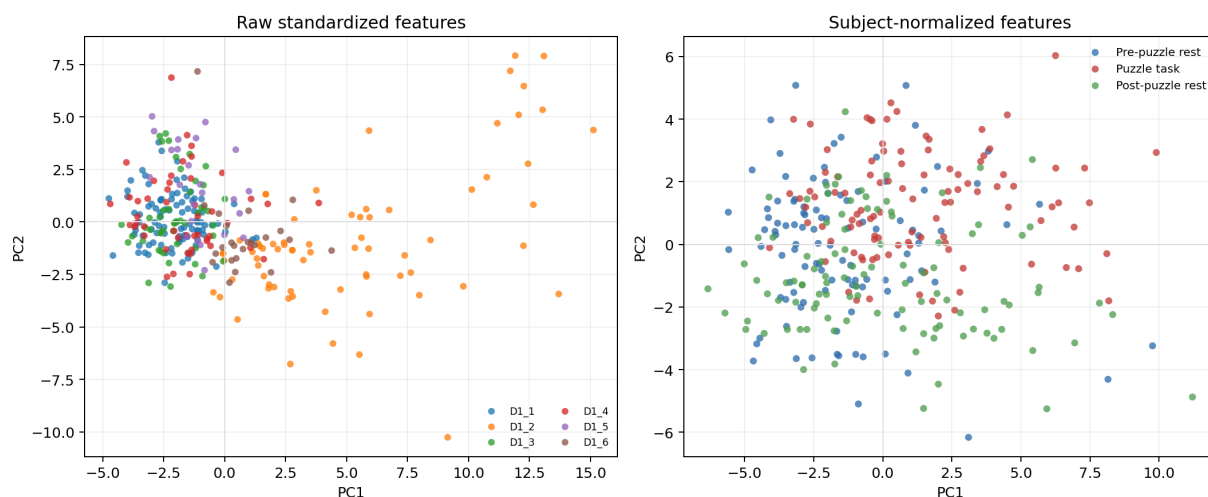


Figure 1. Left: PCA on raw standardized features, colored by cohort. Right: PCA on subject-normalized features, colored by phase. Raw features show clear cohort structure, while subject normalization reduces this baseline effect but does not produce clean phase separation.

Results: What Explains the Representation?

The raw scaled PCA representation showed that the first five PCs explained 58.4 percent of total feature variance. However, the dominant structure was not phase-related. Individual identity explained 64.5 percent of the weighted variance in PC1-PC5, and cohort explained 32.8 percent. Experimental phase explained only 3.0 percent, and puzzler role explained 0.5 percent.

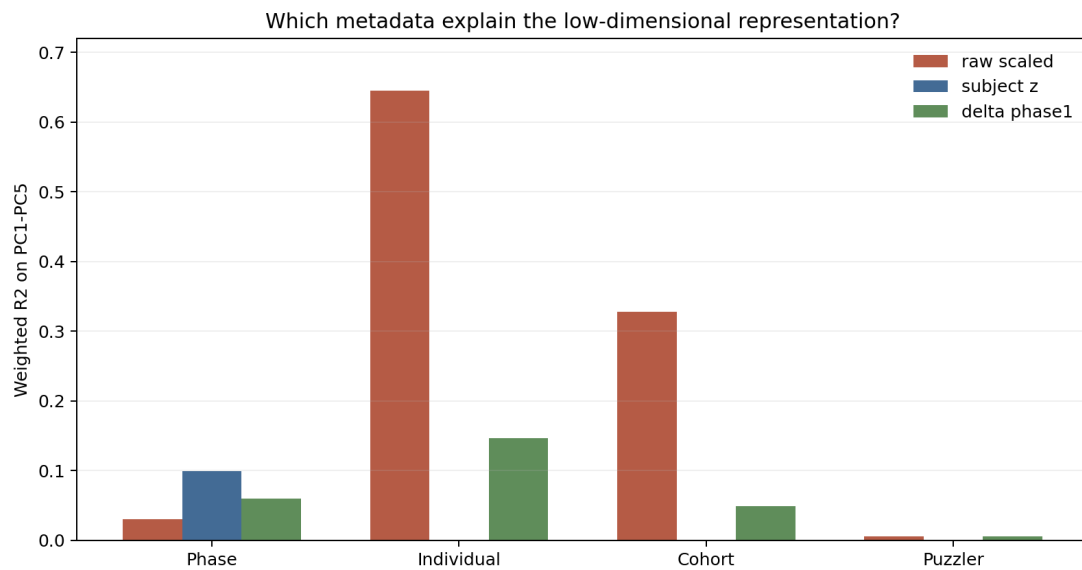


Figure 2. Weighted R2 of metadata variables for PC1-PC5. Subject normalization removes static individual and cohort effects and increases the relative contribution of phase, but the phase signal remains weak.

Pipeline	PC1	PC2	PC1-PC5	Silhouette	ARI	NMI
Raw scaled	0.278	0.100	0.584	0.320	-0.003	0.004
Subject-wise z	0.203	0.091	0.514	0.162	0.075	0.079
Phase1 delta	0.163	0.097	0.493	0.442	0.051	0.129

Table 1. PCA and KMeans summary for the three preprocessing pipelines.

Results: Latent State Discovery

KMeans clustering on the first five PCs did not recover the three phases reliably. For raw scaled features, the ARI was approximately zero (-0.003), indicating no meaningful agreement with phase labels. Subject-wise normalization improved ARI to 0.075, and phase1 baseline subtraction gave ARI 0.051. These values are still low and indicate weak alignment.



Figure 3. KMeans clusters from the subject-normalized PCA representation. The clusters contain mixtures of phases rather than one cluster per experimental phase.

Sanity Checks

The one-class SVM trained on phase1 resting data did not successfully flag phase2 as anomalous under leave-one-subject-out validation. With $\nu=0.10$, phase2-versus-phase1 AUC was 0.402 and balanced accuracy was 0.404. In contrast, the supervised logistic regression sanity check for phase2 versus phase1 achieved AUC 0.779, balanced accuracy 0.692, and F1 score 0.701. This means the data are not completely uninformative, but the signal is subtle and not the dominant source of variance.

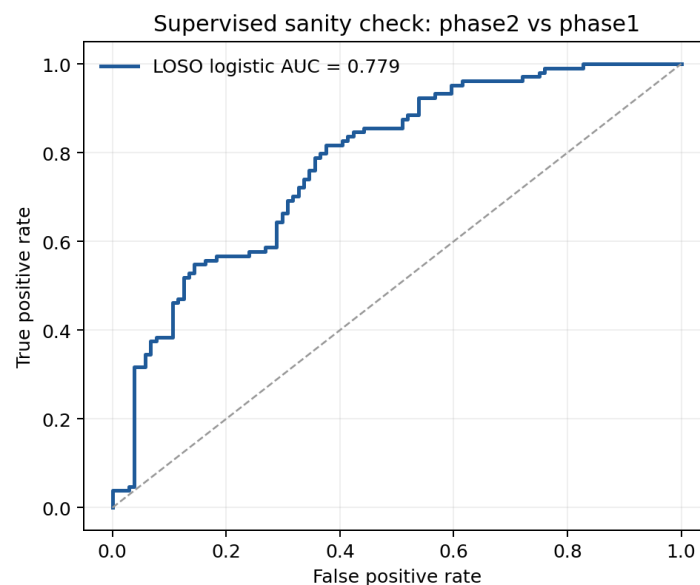


Figure 4. Supervised sanity check for phase2 versus phase1 using leave-one-subject-out logistic regression.

Discussion

The main conclusion is that individual baselines are the dominant source of variation in the raw wearable biosignal features. This is expected for HR, TEMP, and EDA because absolute levels differ substantially across people and recording sessions. Without normalization, a latent representation mostly captures who was measured and when the data were collected, not what experimental state the person was in.

Subject-wise normalization is therefore not an optional preprocessing detail; it changes the scientific interpretation of the representation. It removes static individual and cohort effects and increases the amount of phase-related structure. However, the improvement is limited. The phases still overlap in PCA space, and KMeans clusters remain mixtures of pre-rest, puzzle, and post-rest observations.

The supervised sanity check prevents over-interpreting the negative clustering result. If even a supervised LOSO classifier failed, we might conclude that the feature set contains almost no phase information. Instead, logistic regression achieved AUC 0.779, suggesting that phase information exists but is subtle and distributed across features. PCA may fail because the stress signal is not the largest source of variance, while PCA explicitly prioritizes high-variance directions.

Limitations

The sample size is small, with 26 individuals and only 12 observations per individual. The feature matrix uses summary statistics over phases and may discard important temporal dynamics. Cohort effects are strong, and the cohorts were collected at different times and contexts. Self-reported frustration is subjective and does not necessarily align one-to-one with physiological arousal. Finally, KMeans assumes roughly spherical clusters and may be poorly matched to biosignal manifolds.

Conclusion

The EmoPairCompete biosignals contain weak but nonzero phase-related information, but the dominant structure in the raw data is individual and cohort variability. Subject-wise normalization improves the representation, yet unsupervised PCA plus KMeans does not reliably discover latent states corresponding to pre-puzzle rest, puzzle stress, and post-puzzle recovery. Wearable biosignal analysis therefore requires baseline-aware preprocessing and subject-level validation before latent emotional clusters can be interpreted as meaningful stress states.

Reproducibility

The analysis was implemented in Python using median imputation, standardization, PCA, KMeans, one-class SVM, logistic regression, and leave-one-subject-out validation. The script `submission/case2_analysis.py` reads `HR_data_2.zip`, generates all figures, and exports the numerical result tables used in this report.

GenAI Acknowledgement

Generative AI was used to help structure the report, draft explanatory text, and organize the computational pipeline. All reported numerical results were produced by running the accompanying analysis script on the provided dataset, and the final interpretation was checked against those outputs.

References

- [1] Das, S., Lund, N. L., Gonzalez, C. R., Lonfeldt, N. N., and Clemmensen, L. H. EmoPairCompete - physiological signals dataset for emotion and frustration assessment under team and competitive behaviors. ICLR Workshop on Learning from Time Series for Health, 2024.
- [2] Thompson, E. R. Development and validation of an internationally reliable short-form of the Positive and Negative Affect Schedule (PANAS). *Journal of Cross-Cultural Psychology*, 38(2), 227-242, 2007.